CS341 Task Tracker – User Guide

English | CS341 Project | Java desktop application

1. Requirements

Install Java JDK 11 or newer and Maven 3.8 or newer. Use a normal graphical desktop on Windows, macOS or Linux. The first Maven build downloads dependencies; the application itself uses a local SQLite file and requires no database server.

2. Build and Launch

From the project directory:

```
mvn -B clean verify
java -jar target/task-tracker-1.0.0.jar
```

For Windows PowerShell, use `target\task-tracker-1.0.0.jar`. The database is created as `tasks.db` in the working directory. A custom path is supported with `-Dtasktracker.db=/path/to/tasks.db` before `-jar`.

3. Accounts and Tasks

1. Select **Create account**, choose a username of 3–30 ASCII letters/digits/underscores and a password of at least 8 characters.
2. Sign in. Use **New task** to enter title, description, registered assignee, optional priority and deadline in YYYY-MM-DD format.
3. Select a task to edit it, mark it complete/reopen it, or delete it. The creator can edit/delete; both creator and assignee can change completion.
4. Use **Filter** (All, Open, Completed, High priority) or click table headers to sort the tasks.
5. Log out and sign in as another account to simulate collaboration within the same local database.

4. Common Problems

Problem	Resolution
Maven not found	Install Maven and verify `mvn -version`.
SQLite driver missing	Build the Maven-shaded executable JAR rather than launching raw class files.
No application window	Use a non-headless graphical desktop.
Edit or delete denied	These actions require the creator account.
Invalid date	Enter a real calendar date in YYYY-MM-DD format.

5. Data and Security

The application does not implement automatic backups or encrypted storage. Back up `tasks.db` manually after closing the application, and protect the local file with operating-system permissions. Passwords are not stored as plaintext.